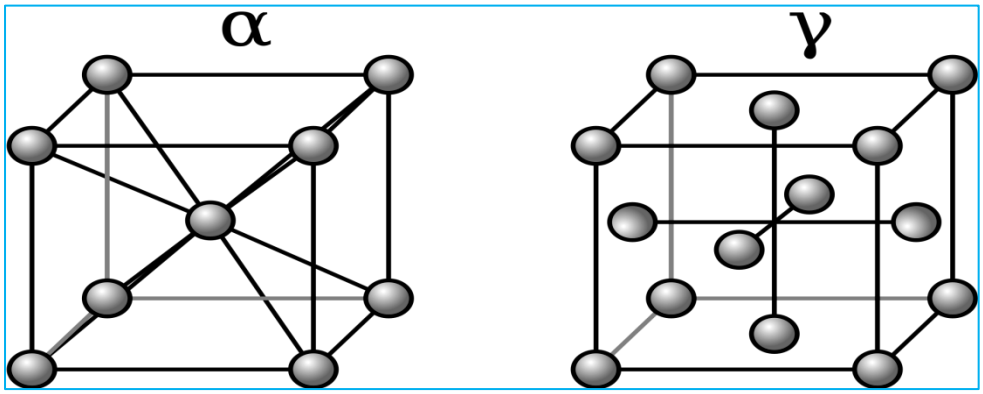
1. INTRODUCTION

1.1. Background of the study

Iron has a chemical symbol Fe, which has 26 atomic numbers and 55.845u atomic mass. Fe in a 3d Transition Metal with a ground state electronic configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$=[Ar]$3d^6 4s^2$. Iron (Fe) has multiple crystal structure depending on its temperature and pressure: bcc, fcc and hcp. But Pure iron can have two different crystal structures as its temperature is increased from room temperature to its melting point. At room temperature to 912°c it is body-centered cubic (**α-Fe**), between 912°c and 1394 ^{o}C it is face-centered cubic (γ-Fe), and between 1394 and its melting point at 1538 ^{o}C it returns to body-centered cubic[1]. bcc and fcc structures are usually referred to as "ferrite" and "austenite" respectively. The boiling point of iron is about 3300K. A third form of iron, which has the hexagonal close packed (hcp) structure, can be stabilized under high pressures[2].

Fe is Tungsten structured and crystallizes in the cubic I$\bar{m}$3m space group. Fe is



bonded in a distorted body-centered cubic geometry to eight equivalent Fe atoms. All Fe-Fe bond lengths are 2.48 Å.

Iron, as commonly available, nearly always contains small amounts of carbon, which are picked up from the coke during smelting. At atmospheric pressure, three allotropic forms of iron exist, depending on temperature: alpha iron (α-Fe, ferrite), gamma iron (γ-Fe, austenite), and delta iron (δ-Fe). The lattice parameter is 0.3571 nm for FCC iron and 0.2866 nm for BCC iron. Assume that carbon atoms have a radius of 0.071 nm[3].

The word ferroalloys refers to an alloy of iron containing a significant proportion of one or more other elements like silicon, manganese, chromium, aluminum, or

titanium. The main applications of ferroalloys occur in the steelmaking process. They are added to steel to improve properties like strength, ductility, and fatigue or corrosion resistance. Additionally, ferroalloys can have several other tasks, such as in refining, deoxidation, modification, and control of nonmetallic inclusions and precipitates. Stainless steel is an alloy of iron, chromium, and nickel. Stainless steel is formed by alloying iron with nickel and chromium in order to prevent rusting of iron Silicon. The element silicon dissolves into iron to make stronger steel. Silicon is often used as a deoxidizer in something such as rolled steel. This makes the alloy have a higher tensile strength, as well as forgeability and hardness. Iron is used in manufacture of steel, used in civil engineering (reinforced concrete, girders etc.) and in manufacturing. In the Earth's crust iron is found mainly as minerals of iron oxide such as hematite, magnetite, goethite and limonite. The minerals that are mostly used as ore for making iron are hematite (Fe2O3) and magnetite (Fe3O4). this study is aimed at the first principle calculation of iron using Density Functional Theory and Quantum Espresso packages as tools.

1.2 Statement of the Problem

The electronic and structural properties of single particle is easy to calculate but it is known that many body problems are difficult to solve. That is, the state of motion cannot be solved analytically for systems in which three or more distinct electrons interact[4,5]. In particular Hatree-Fock theory and its source are based on the complicated many-electron wave function, where as many-body wave function is dependent on 3N variables, three special variables for each of the N electrons. Now a days, there is more applicability in the fundamental principle of density functional theory in any properties of a system of many interacting particles can be viewed as a functional of the ground states of density, $no(r)$, which in principle determine all information in the many-body wave function for the ground states and all excited states. Schrodinger equation is setting of practical values and many applicable rapidly in research in chemistry, physics. Many researchers have been studying the electronic and structural properties of Fe as function of cohesive energy, bulk modulus and density of states. However we are interested to study the electronic and structural

properties of Iron per atom as a function of energy cutoff, lattice constant, k-point sampling and smearing for k-point sampling with the help of density functional theory.

1.3 Objectives

The study has the following general and specific objectives.

1.3.1. General objective

The general objective is to study the role of crystal systems and magnetic structures on the

electronic properties of 3d transition metal of Fe with respect to density functional theory.

1.3.2. Specific Objectives

Specifically, the objectives of the study are:

- To calculate the total minimum energy of Fe per atom with respect to ECUT-OFF energy & to K-POINT sampling.
- To calculate the total magnetization of Fe per atom with respect to degauss.
- To calculate the total stress of Fe per atom with respect to degauss.
- To calculate the total minimum energy of Fe per atom with respect to different smearing/degauss for k-point sampling.

1.4. Significance of the study

- Gives a basic understanding of electronic structures for current and prospective undergraduate and graduate students for further research
- Helps early-stage researchers who are not familiar with the Quantum ESPRESSO package.
- To get an insight into the quantities when compared to the experimental values.
- To understand different properties of Fe which in turn helps its application in modern

electronics and optical devices.

- The computational seeks to gain understand of electronic and structural properties of electron principally through use and analysis of mathematical models on high performance computer and to create new knowledge.

1.5. Thesis Organization

This Thesis was organized into five chapters, In chapter one we discuss the back ground information about Iron structure, Chapter two presents about related literature review with basic information on Iron. its properties, and its application, Chapter three deals with the computational method. It started with density functional theory and with Hohenberg Kohn theorem, algorithm implementation, plane wave expansion, and Bloch theorem. The results and discussion part of the Thesis was described in Chapter 4. Finally chapter five presents the summary and future works in related to this work.

2. RELATED LITERATURE REVIEW

2.1. Crystal Structures

Metals are often found and fabricated in a number of different crystal structures. The metal's properties can be manipulated to meet the demands of the requirements imposed by the user. For example, steel used for a kitchen knife will have very different properties (inelastic) than steel used as girders in the construction of an office building (elastic). At ordinary temperatures, iron has a body-centered cubic (bcc) crystal structure. If the temperature of the iron is raised above 910° C, it changes to a face-centered cubic (fcc) crystal structure. Impurities mixed into the crystal (alloys) also affect the metal's properties and characteristics. For example, addition of a small amount of carbon in the iron forms steel and greatly affects its structure and properties. To demonstrate these types of changes and manipulations, we can use a simple hairpin to gain a better understanding of the metallurgy of iron. Iron has two different crystal structures at atmospheric pressure: the body centered cubic (bcc) and the face centered cubic (fcc). In the ground state the bcc α-phase is stable, and at the temperature T=1184 K (A3 point), α-Fe transforms into fcc α-Fe, which is stable up to 1665 K (A4 point). Above this temperature, iron transforms back into the bcc phase (σ-Fe), which remains stable up to the melting temperature Tm =1809 K. Since α and σ-Fe are isomorphic, the two are usually not distinguished when referred to bcc Fe. Steels with bcc and fcc structures are usually referred to as "ferrite" and "austenite" respectively [6]. The boiling point of iron is about 3300K. A third form of iron, which has the hexagonal close packed (hcp) structure, can be stabilized under high pressures Austenite, ferrite, and martensite are different microstructural phases in steel and other iron-based alloys, each with distinct properties due to their atomic arrangements and compositions:

1. Austenite (γ-iron)

Structure: Face-Centered Cubic (FCC) crystal structure.

Formed: At high temperatures, above the critical temperature known as the A3 point (around 912°C for pure iron), where iron is Austenite.

Properties Soft ductile, and non-magnetic Can dissolve more carbon than

ferrite, which leads to the formation of various micro structures upon cooling.

Transformation: On cooling, austenite can transform into pearlite, bainite, or martensite depending on the rate of cooling.

2. Ferrite (α-iron)

Structure: Body-Centered Cubic (BCC) crystal structure.

Formed: At room temperature or slightly below 912°C (A3 point) where iron is ferritic.

There's also delta ferrite (δ-iron), which forms at temperatures just below the melting point of iron and has the same BCC structure.

Properties:Soft, ductile, and magnetic.

Lower carbon solubility compared to austenite; typically contains very little carbon in solid solution.

Transformation: Ferrite can form directly from austenite during slow cooling or be present in a mixed micro structure with pearlite.

3. Martensite

Structure: Body-Centered Tetragonal (BCT) in high carbon steels, but can be close to BCC in low carbon steels due to the distortion caused by carbon atoms. Formed: By very rapid cooling (quenching) of austenite, which does not allow enough time for diffusion, thus trapping carbon atoms within the iron matrix.

Properties:

Very hard and brittle due to the supersaturated carbon trapped in the iron lattice, causing lattice distortion.Not a state of equilibrium; it's a metastable phase. Martensite can be tempered to reduce brittleness and improve toughness. Transformation: The transformation from austenite to martensite is diffusionless (athermal), where the crystal structure changes without the movement of atoms other than the iron atoms themselves adjusting their positions.The standard enthalpy of formation of Fe is 0.00 kJ/mol at 25°C, and 75.90 kJ/mol at 1600°C [7].

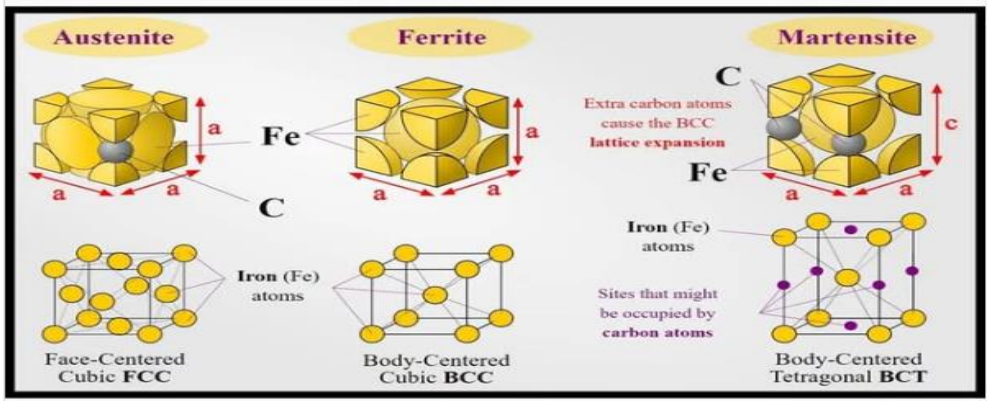


Figure 1:Crystalline structures and the slip system of Iron: BCC and FCC of Fe.

2.2. Properties

2.2.1. Electronic properties

The electronic properties of Iron (Fe) are characterized by its electronic configuration, electronic band structure, oxidation state and in different chemical environments. Here are some key points about electronic properties of Iron.

Iron-based materials are roughly classified into pure iron, steel, and cast iron according to carbon content. Steel in particular is used in many industrial fields as special stainless steel as well as lead frames and other electronic materials by adding various alloy elements, such as chromium and nickel, to it. Metallic iron can crystallize in three different crystal structure:base centered cubic (bcc) , face centered cubic (fcc) and hexagonal closed-packed (hcp). At pressures above approximately 10-13 GPa and temperatures up to around 700 K, α-iron changes into a hexagonal close-packed (hcp) structure, which is also known as ε-iron or hexaferrum;the higher-temperature γ-phase also changes into ε-iron, but generally requires far higher pressures as temperature increases[8]. During the past years several groups have concentrated their efforts on the preparation and characterization of nano structured systems derived from these bulk phases, mainly due to their interesting magnetic and catalytic properties. These studies have shown a strong correlation between crystal structure and magnetic properties in Fe-based materials.It is the arrangement of electrons in the increasing order of energy in the shells of an atom. Iron is a transition metal and the atomic symbol of Iron is(Fe). The atomic number of Iron 26. The Electronic configuration of Fe: $[Ar]3d^6 4s^2$. The commonly written electron configuration using the Aufbau principle has the 3d and 4s sublevels reversed along

with the erroneous thinking that 4s is somehow lower in energy than 3d. That is incorrect. The 4s sublevel is higher in energy than the 3d with elements with atomic numbers of 21 and above. Therefore, the 4s is written last in the electron configuration. This configuration indicates that iron has a total of 26 electrons, with six of them occupying the d orbital. The partially filled d orbitals contribute to iron s interesting electronic and magnetic properties.

Iron has two valence electrons in the outermost shell ($4s^2$), and six valence electrons in the d orbital ($3d^6$). These valence electrons play a significant role in iron reactivity, Chemical Bonding,Oxidation States,Magnetism and the other.electronic band structure is the foundation of solid-state physics, describing how electrons behave in crystalline materials. It explains the formation of energy bands and gaps, which determine a material's electrical properties. Understanding band structure is crucial for developing electronic devices and materials[9]. In solid-state physics, the electronic band structure of iron describes the range of energy levels that electrons can occupy within the material, as well as the energy gaps where electrons cannot exist. This band structure arises due to the periodic arrangement of atoms in the crystal lattice, which causes the atomic orbitals to merge and form continuous energy bands.

2.2.2. Magnetic properties

The magnetic properties of surfaces are governed by the behavior of the electron system. Several previous calculations for individual iron surfaces allowed to predict the enhancement of the surface magnetic moments[10]. However, being performed within different approximations (relaxed and unrelaxed slabs, different energy functionals) they are unsatisfactory for predicting their correlation with the atomic coordination at the surface[11].

Here, we report comprehensive calculations of structural, electronic, and magnetic properties for the seven most densely packed iron surfaces, addressing the behavior of the above-mentioned quantities. We systematically examine the relationships between all these properties and the surface roughness, taking into account a complete relaxation of the atomic positions. To the best of our knowledge, such a

comprehensive study of all these characteristics for several low-index surfaces of a single 3d transition metal, has not been reported so far[12].

The Stoner theory of itinerant ferromagnetism was used in calculating the magnetic properties of iron within the self-consistent linear muffin-tin orbital approach. The Stoner exchange parameter, *I*, was found from the linear-response theory as a function of volume and then adjusted by a constant enhancement factor determined by fitting the equilibrium atomic volume of the ferromagnetic (FM) bcc phase. No other adjustments of any quantities were performed. The calculations revealed a somewhat unusual behavior of phases. The nonmagnetic (NM) bcc phase is unstable with respect to spontaneous magnetization[13 ,14,15]. Moreover, the enthalpy of the NM phase exhibits a monotonic decrease with c/a along the so-called Bain deformation path, the bcc modification being unstable with respect to the shear deformation. Ferromagnetism stabilizes the bcc phase. However, the FM fcc phase is unstable with respect to the shear deformation. The enthalpy curve along the deformation path has a cusp corresponding to a first-order phase transition between FM and NM states. The FM contributions to the energy versus atomic volume for different c/a valves are tabulated.

2.3. Applications

Iron alloys have a high resistance to corrosion and rust formation. It may also be easily bent to create a variety of shapes. This is why iron is used in the majority of the things that you see around the house. Wrought iron is being used in a variety of outdoor goods to improve the appearance of the garden and terrace. Cast iron is another essential type of iron that is used to make pans, pots, ovens, skillets, and trays. Additionally, cast iron contains a heating ingredient that is ideal for cooking products, kitchenware, and other accessories. Furthermore, stainless steel is used in kitchen utensils and many electrical gadgets. Bridges, skyscrapers, buildings, and other types of construction projects would be completed if their structures did not include iron. Iron is utilized in the fabrication of metal items, buildings, the manufacture of toys and goods, and in the manufacturing of transportation equipment. Ferric chloride, an iron compound, is used in wastewater treatment plants to remove pollutants from water.

Surgical equipment is made from stainless surgical steel, often known as inox steel. As modern reagents, nitrates and iron chloride are used. The fungicide makes use of iron sulphate. They are commonly used in the construction of large boats, automobiles, other machine devices, and machine parts[16].

3. COMPUTATIONAL METHOD

3.1. Density Functional Theory (DFT)

Density functional theory (DFT) is a quantum-mechanical atomistic simulation method to compute a wide variety of properties of almost any kind of atomic system: molecules, crystals, surfaces, and even electronic devices when combined with non-equilibrium Green's functions (NEGF).

DFT belongs to the family of first principles (ab-initio) methods, so named because they can predict material properties for unknown systems without any experimental input. Among these, DFT has earned popularity due to the relatively low computational effort required.

The DFT approach is widely applied in organic and inorganic chemistry, materials sciences like metallurgy or ceramics, and for electronic materials, to just name a few areas.[17]. Density Functional Theory (DFT) is one of the most widely used methods for 'ab initio" calculations of the structure of atoms, molecules, crystals, surfaces, and their interactions. Unfortunately, the customary introduction to DFT is often considered too lengthy to be included in various curricula. An alternative introduction to DFT is presented here, drawing on ideas which are well-known from thermodynamics, especially the idea of switching between different independent variables. The central theme of DFT, i.e. the notion that it is possible and beneficial to replace the dependence on the external potential v(r) by a dependence on the density distribution n(r), is presented as a straightforward generalization of the familiar Legendre transform from the chemical potential (\mu) to the number of particles (N). This approach is used here to introduce the Hohenberg-Kohn energy functional and to obtain the corresponding theorems, using classical nonuniform fluids as simple examples. The energy functional for electronic systems is considered next, and the Kohn-Sham equations are derived. The exchange-correlation part of this functional is discussed, including both the local density approximation to it, and its formally exact expression in terms of the exchange-correlation hole. A very brief survey of various applications and extensions is included.[18]

DFT is an ab initio method, meaning that it does not use any experimental results on chemical bonding. It works with the electron density. There are significant advantages to a computational theory based on electron densities.
DFT is a successful approach (a reliable tool) for studying or finding solutions to the fundamental equation of the quantum behavior of atoms and molecules at low energies.

DFT is the many body theory based on the idea of using only the density as the basic variable for describing many electron systems. It is widely used in condensed matter theory and computational material science for the calculation of electronic, magnetic and structural properties of solids.

Density functional theory is an extremely successful approach for solving many body problems' i.e., it is an exact reformulation of many-body quantum mechanics in terms of the probability density rather than the wave function. The following are among approximation approaches of density functional theory [19].Density functional theory (DFT) is a quantum mechanical technique used in physics and chemistry to investigate the structural and electronic properties of many body systems [20]. It is a successful approach to the fundamental equation that describe the quantum behavior of atoms and molecules to simple crystals and complex extended system [20, 21]. It allows one to replace the complicated N-electron wave function $\Psi(r1, r2, ..rN)$ and associated with Schr¨odinger equation by the much simpler electron density $n(r)$ [22]. Density functional theory as a good modeling method has become a common tool in first-principle calculations aimed at molecular and condensed matter systems [23]. Traditional methods in electronic structure theory, in particular Hatree-Fock theory(HFT) and its descendants are based on the complicated many-electron wave function. The main objective of DFT is to replace the many-body electronic wave function with the electronic density as the basis quantity, where as many-body wave function is dependent on 3N variables, three special variables for each of the N electrons. The density is only a function of three variables and is a simpler quantity to deal with both conceptually and practically [24]. The modern formulation of density functional theory originated in a famous paper written by P. Hohenberg and W. Kohn with their two fundamental theorems. In 1965 the major milestone in the development

of DFT was introduced. They gave the proofs of these theorems by showing that DFT was an exact theory in same sense as the wave function theory [25, 26, 27].

3.1.1 The Hohenberg-Kohn theorem

The foundation of the DFT method is the Hohenberg-Kohn theorem, which states that for each given electronic density $\rho(r)$, there is one and only one corresponding potential. All properties of the many body system are determined by ground state density. The H-K theorem implies that the ground- state for any system can be determined by varying the charge density until the global minimum in the energy functional is found. There are two Hohenberg-Kohn Theorem's[25].

3.1.1.1 Hohenberg-Kohn Theorem 1

For any system of interacting particles in an external potential $V_{ext}(r)$, the potential $V_{ext}(r)$ is determined uniquely, not including a constant, ground state particle density $n0(r)$ [28].

- Uniqueness of the Ground State Density: For a system of interacting electrons in an external potential V(r) , the ground state density ρ(r) uniquely determines the external potential V(r) (up to a constant). This means that if two different external potentials yield the same ground state electron density, they must be identical (except for an additive constant).

- Existence of a Ground State: The theorem also asserts that there exists a unique ground state density for any given external potential that is sufficiently well behaved.

This theorem implies that all properties of the ground state of a many-electron system can be derived from its electron density rather than from the many-body wave function.

3.1.1.2 Hohenberg-Kohn Theorem 2

The second HK theorem defines energy functional for the system and proves that the correct ground state electron density minimizes this energy functional [29]. The energy functional of the density E[n] is:

$$E[n] = \int dr V_{ext}(r) n(r) + F[n],$$

where $F[n]$ is a universal functional of the density and incorporates the kinetic and the potential energy. Once the external potential $V_{ext}(r)$ has been fixed, the energy functional E[n] has its , the ground state energy E_0, at the physical ground state density $n_0(r)$:

$E_0 = E[n0]$.

The Hohenberg-Kohn (HK) theorems have the limited purpose to prove that a universal functional of the electron density exists; they do not derive its actual ex pression. A direct minimization of the functional is usually not applicable, because no good expression for the kinetic energy as a functional of n is known, except for simple metals. The Kohn-Sham (KS) scheme, a reformulation of the theory based on the KS orbitals instead of the mere density, is the starting-point of most of the actual calculations.

Hohenberg-Kohn Theorem 2 provides a powerful framework for calculating the properties of many-electron systems by focusing on the electron density and establishing a variational principle that leads to the determination of ground state energies and densities. This theorem is foundational to the practical implementation of density functional theory (DFT).

3.1.3 EXCHANGE CORRELATION APPROXIMATION

The exchange-correlation energy E_{xc} of a many- electron system is the quantity of DFT. In context of Kohn-Sham theory, E_{xc} is defined as a functional of the electron density ρ. In K-S expression the total electronic energy $E[\rho]$ is;

$E[\rho] = Ts[\rho] + V[\rho] + W[\rho] + Exc$

Where; Ts - is the kinetic energy of a non-interacting particle system with density ρ, V is the energy of electron-nuclear attraction, W H is the coulomb or Hartree energy and E_{xc} is the exchange-correlation energy. An accurate values of the exchange and correlation energies obtained for chemically interacting systems are essential for analysis of the effect of electron correlation with the Kohn-sham theory and in order to test and calibrate various DFT approximation (Local Density Approximation and Generalized Gradient Approximation).

A. Local-density approximations (LDA)

LDA: are a class of approximations to the exchange–correlation (XC) energy functional in density functional theory (DFT) that depend solely upon the value of the electronic density at each point in space (and not, for example, derivatives of the density or the Kohn–Sham orbitals). Many approaches can yield local approximations to the XC energy. However, overwhelmingly successful local approximations are those that have been derived from the homogeneous electron gas (HEG) model. In this regard, LDA is generally synonymous with functionals based on the HEG approximation, which are then applied to realistic systems (molecules and solids).

The importance of these computational studies stems from the system complexities which bring about high sensitivity to synthesis parameters necessitating first-principles based analysis. The prediction of Fermi level and band structure in doped semiconducting oxides is often carried out using LDA incorporated into simulation packages such as CASTEP and DMol3.[2] However an underestimation in Band gap values often associated with LDA and GGA approximations may lead to false predictions of impurity mediated conductivity and/or carrier mediated magnetism in such systems.[3] Starting in 1998, the application of the Rayleigh theorem for eigenvalues has led to mostly accurate, calculated band gaps of materials, using LDA potentials.[4][1] A misunderstanding of the second theorem of DFT appears to explain most of the underestimation of band gap by LDA and GGA calculations, as explained in the description of density functional theory, in connection with the statements of the two theorems of DFT.

B / The generalized gradient approximation (GGA)

This functional (GGA) depends on the local electron density as the spatial variation of the electron density that is represented by density gradient. The idea behind this functional was to improve the approximation of LDA by considering not only the electron density, but also the local gradient of that density [31,32]. The GGA functional can be written as;

$$E_{xc}^{GGA}[n] = \int n(r)\, E_{xc}[n(r)\, \nabla_n]\, d\vec{r}$$

The $E_{xc}^{GGA}[n]$ is the exchange correlation energy per particle of an electron gas. The

GGA gives better total energies. When a bond between two atoms is broken, the surface is increased. In GGA, this bond-breaking process is more favorable than in LDA and hence bond is weakened. Thus the GGA cures the over binding error of the LDA. These gradient corrections greatly improved the bond energies and made density functional theory useful also for chemists. The most widely distributed GGA functional is the Perdew Burke-Ernzerhof (PBE) functional [31,32,33].

3.2 plane wave [34]

3.3 K-points sampling

The solution that is used most widely was developed by Monkhorst and Pack in 1976. Using these methods, one can obtain an accurate approximation for the electronic potential and the total energy of an insulator or semiconductor by calculating the electronic states at a very small number of k-points. The electronic potential and total energy are more difficult to calculate if the system is metallic because a dense set of k-points is required to define the Fermi surface precisely. The magnitude 9of any error in the total energy due to inadequacy of the k-points sampling can always be reduced by using a denser set of k-points. The computed total energy will converge as the density of k-points increases, and the error due to the k-point sampling approaches zero.

In principle, a converged electronic potential and total energy can always be obtained provided that the computational time is available to calculate the electronic wave functions at sufficiently dense set of k-points. The computational cost of performing a very dense sampling of k-space can be significantly reduced by using the k-point total energy method [35-36].

3.4 PseudoPotential

From the earliest developments of plane-wave methods, it was clear that there could be great advantages in calculations that approximated the properties of core electrons in a way that could reduce the number of plane waves necessary in a calculation.

The most important approach to reducing the computational burden due to core electrons is to use pseudo-potentials. Conceptually, a pseudo-potential replaces the electron density from a chosen set of core electrons with a smoothed density chosen to match various important physical and mathematical properties of the true ion core. The properties of the core electrons are then fixed in this approximate fashion in all

subsequent calculations; this is the frozen core approximation. Calculations that do not include a frozen core are called all-electron calculations, and they are used much less widely than frozen core methods. A pseudopotential is developed by considering an isolated cell of one element, but the resulting pseudopotential can then be used reliably for calculations that place this cell in any chemical environment without further adjustment of the pseudo-potential. This desirable property is referred to as the transferability of the pseudo-potential. The details of a particular pseudopotential define a energy cutoff that should be used in calculations including cells associated with that pseudopotential. Pseudopotentials requiring high cutoff energies are said to be hard, while more computationally efficient pseudo-potentials with low cutoff energies are soft. The most widely used method of defining pseudo-potentials is based on work by Vanderbilt; these are the ultrasoft pseudopotentials (USPPs). As their name suggests, these pseudopotentials require substantially lower cutoff energies than alternative approaches [37]

3.5 Quantum ESPRESSO code

Quantum ESPRESSO (Quantum open-source Package for Research in Electronic Structure, Simulation, and Optimisation) is a suite of applications for ab-initio electronic structure calculations using plane waves and pseudo-potentials.

Quantum ESPRESSO is an integrated module of computer codes for electronic and structural

calculations and materials modeling depending on the framework of the

- ✓ DFT
- ✓ plane wave basis sets (PW), and
- ✓ Pseudo-potentials to represent the electron-ion interactions.

It is free, open-source software distributed under the terms of the GNU General Public Licence (GPL). Also, several sets of Monkshood-pack k-point grid samplings were tested. Plane waves are easy lattice-periodic basis functions but in general not-possible to expand the crystal wave function in plane waves because of the strong oscillation near the cores (almost infinitely many plane waves) would be required. The pseudopotential method avoids the problem. A very famous one is the ultra-soft pseudopotential. The ultra-soft pseudo potential is constructed to be smoother, which has the advantage that an even smaller amount of plane waves are needed

for the expansion. A popular implementation of the ultra-soft pseudopotential method is the open-source plane-wave self-consistent field (PWSCF). The PWSCF or the iterative approach to self-consistency using different techniques in the framework of the plane wave pseudopotential method with regards to the ultra-soft pseudo potentials are implemented.

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